Where Should New Fire Stations Go?

Hot spot analysis of emergency medical calls in Fort Worth, Texas

Davina Whata, Michigan Technological University, January 2026

Overview

The Fort Worth Fire Department wanted to know whether emergency medical service (EMS) calls in Battalion 2 tend to cluster. If they do, emergency intensive care units and new stations could be placed near the hot spots. Three spatial statistics were applied to calls from February 2015, each answering a different question, and the results were then used to propose new station locations.

Data and Setup

The analysis used EMS calls for February 2015 (EMS calls only, and all calls for service), the Battalion 2 boundary, active and proposed fire stations, major roads and alarm territories. All layers were in NAD 1983 StatePlane Texas North Central FIPS 4202 (US Feet), except the major roads, which were reprojected to match. Because the existing area field could have been calculated in different units or a different coordinate system, the Battalion 2 area was recalculated in US survey square feet: 520,175,356 sq ft.

1. Are EMS Calls Clustered?

The Average Nearest Neighbor tool compares the average distance from each call to its nearest neighbor with the distance expected if the same number of calls were spread randomly over the same area. The null hypothesis was that EMS calls are randomly distributed across Battalion 2.

Statistic	Value
Observed mean distance	287.18 ft
Expected mean distance	523.24 ft
Nearest neighbor ratio	0.549
z-score	-18.81
p-value	< 0.001

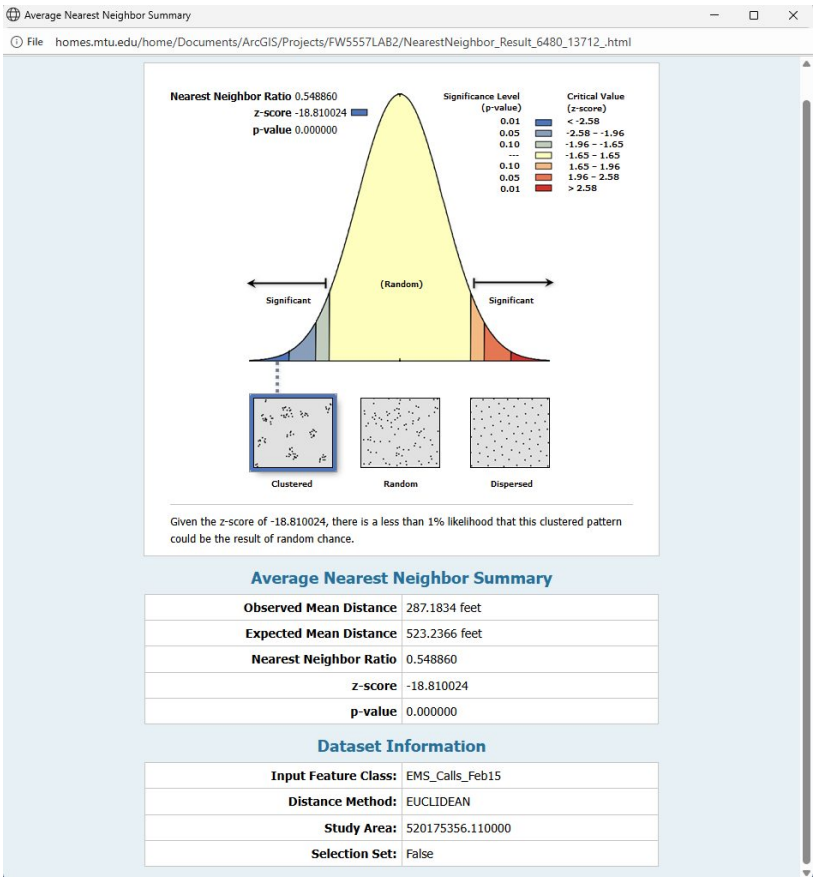


Figure 1. Average Nearest Neighbor result for EMS calls in Battalion 2.

Calls sit on average only about half as far from their nearest neighbor as a random pattern would predict (ratio 0.549). With a z-score of -18.81 and $p < 0.001$, the null hypothesis is rejected: EMS calls are significantly clustered rather than randomly distributed.

2. Do High-Priority Calls Cluster?

Location alone does not show whether the most serious calls cluster. The High/Low Clustering (Getis-Ord General G) tool was therefore run on all calls for service using their priority ranking (FEE, from 1 for a cancelled call to 10 for a rescue or vehicle accident with injuries), with a fixed distance band. The average distance to the 7 nearest neighbors was 1,001 ft, so distances from 801 to 1,201 ft were tested in 100-ft steps to find where clustering peaks.

Distance (ft)	Observed G	z-score	p-value
801	0.033	7.94	< 0.001
901 (peak)	0.037	8.08	< 0.001
1,001	0.039	7.89	< 0.001
1,101	0.041	7.88	< 0.001
1,201	0.043	7.99	< 0.001

At every distance, high-priority calls cluster significantly. The z-score peaks at about 900 ft, the scale at which clustering of high-priority calls is strongest.

3. At What Scale Do Call Densities Cluster?

To look at call density per city block, a 500-ft grid (roughly one block) was overlaid on the calls and the number of calls in each cell was counted. Global Moran's I (zone of indifference, Euclidean distance) was then run on cells containing at least one call, starting at 3,400 ft (at least 7 neighbors) and increasing in 200-ft steps.

Distance (ft)	Moran's I	z-score	p-value
3,400	0.011	1.55	0.121
3,600	0.011	1.64	0.100
3,800	0.011	1.64	0.100
4,000 (peak)	0.016	2.46	0.014
4,200	0.015	2.45	0.014
4,400	0.012	2.10	0.036
4,600	0.010	1.88	0.059

Clustering of call densities becomes significant at 4,000 ft and peaks there (Moran's I = 0.016, z = 2.46, p = 0.014), then weakens at larger distances.

4. Proposing New Stations

The proposed stations were overlaid on the hot spots using the 4,000-ft distance band to check whether they sit in the best areas to serve higher-ranking calls. New proposed stations were then placed on major roads, near the clusters of high-priority calls, and their coverage was mapped alongside the active stations.

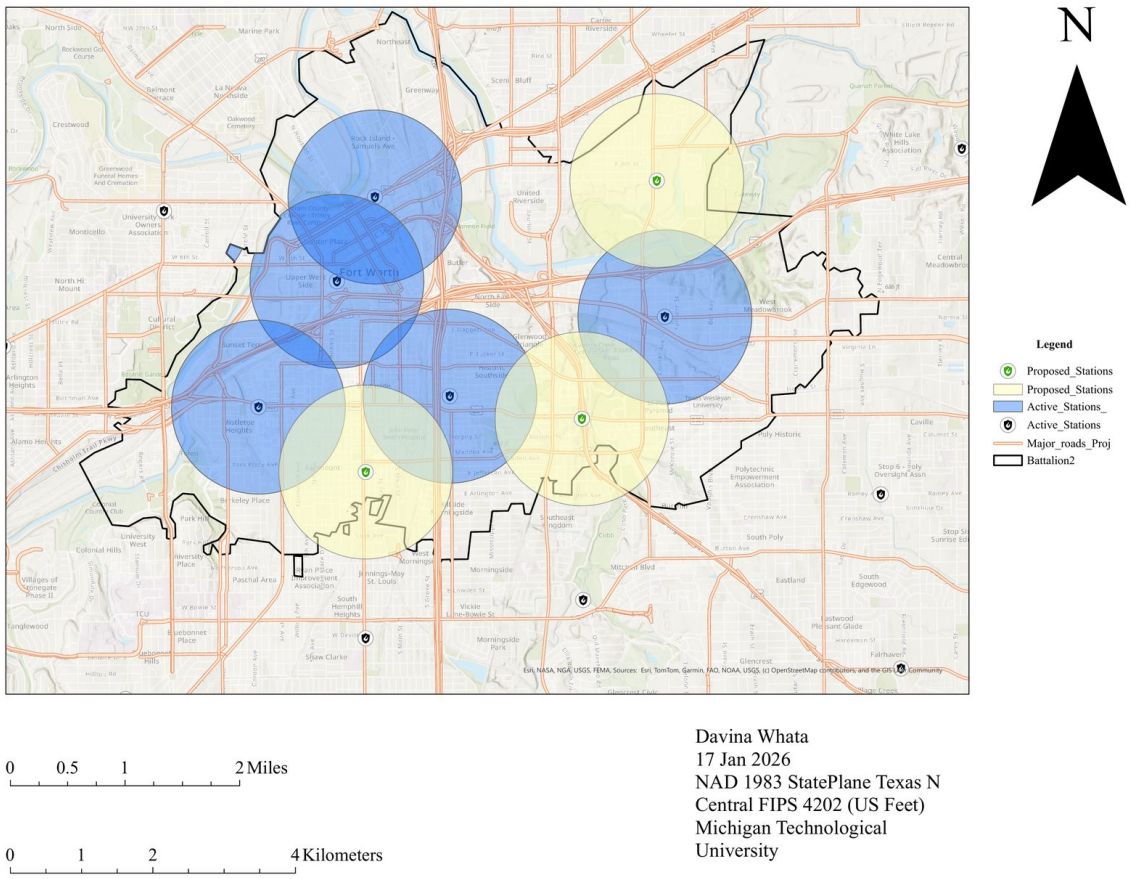


Figure 2. Active (blue) and proposed (yellow) Battalion 2 stations with their coverage areas.

Conclusion

EMS calls in Battalion 2 are strongly clustered, high-priority calls cluster most strongly at around 900 ft, and call densities cluster at around 4,000 ft. Placing new stations on major roads close to these hot spots extends coverage to the areas where serious calls concentrate.